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Sensing the World FYRE Module

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To begin my final project, I worked through the project worksheet with my teammates. We brainstormed multiple ideas and landed on an automatic umbrella system. The purpose of the umbrella is to help people with disabilities who aren't able to open an umbrella themselves when it is raining. Completing the worksheet helped us to gather our plans into one place and feel prepared to start working in the lab. I will use this process of organizing my objectives for future assignments and projects implementing the engineering design process.

At the next class, we started bringing our prototype to life in the Keysight Laboratory. We each worked on one component of the system to be most efficient in creating a functioning prototype. Sam worked on building the model umbrella, Brandon worked on the breadboard configuration, and I worked on creating a detailed prompt for ChatGPT to make the micropython code. As we worked through parts of the process, we gave suggestions to each other and helped work through challenges anyone ran into. We worked very well collaborating as a group and everyone contributed equally to the creation of our umbrella. After a few hours, we had everything set up and ready for an initial test. The first time we ran the code, the LED light worked as intended, but the servo motor was not moving the umbrella up. We explained the issue to ChatGPT and figured out our wires connected to the motor were incorrect. After fixing this, the prototype functioned as we had intended. I was most surprised about the strength of the servo motor. The umbrella was relatively heavy, so we were concerned the motor would not be able to move it up and down. However, it worked perfectly and we didn't have to change the umbrella design.

Our next objective was to improve the design of the prototype itself and make the LED light a more effective indicator. We decided to make a cardboard box to represent a wheelchair

that the umbrella would connect to. We attached the umbrella and the breadboard to the top of the box. Then, I thought about making the LED light a rain intensity indicator instead of just showing if it is raining or not. We generated a second version of the code using ChatGPT, implementing the LED blinking speed to increase for heavier rain and decrease for light rain. With the final code, umbrella, breadboard setup, and wheelchair box, our prototype was complete.

The biggest challenge we encountered when testing our prototype was the inconsistency of our humidity sensor's ADC readings. The readings reflected how well the wires touched the ends of the sensor. It was difficult to keep the wires attached correctly with tape, but with a lot of adjusting we got it to work consistently. If I did a project like this in the future, I would want to research a better method of getting accurate and consistent readings from the humidity sensor.

The final step of the project was creating an accurate and engaging presentation to convey what we did. For the slides, we focused on minimising words by only writing the important points we would discuss. We added pictures of the prototype in its different stages and other visuals from google and our ChatGPT chatlog. I was satisfied with how our presentation came out, and we felt well prepared to speak in front of the class and have everyone clearly understand our process. This part of the module was by far my favorite, and I gained a lot of valuable skills from it. Collaborating with others is the aspect of engineering that I enjoy most, and it was very rewarding to make a successful prototype and presentation with my teammates. I also learned how to use the engineering design process, especially how to work through challenges as a team. I thoroughly enjoyed each part of this project and I hope to continue doing similar work at Lehigh and in my future career.